

Agentic Personalized Meal Planning for Chronic Disease Management

Author/s: Daniel Costa

Abstract:

Chronic diseases are the leading cause of illness, disability, and death in America. Conditions such as diabetes, heart failure, and osteoporosis are strongly influenced by dietary behavior, yet adherence to therapeutic diets remains a major challenge. Existing nutrition solutions primarily focus on calorie tracking, behavioral coaching, or limited-duration clinical counseling, but often fail to help users operationalize dietary guidance into realistic daily meals. Patients frequently struggle with portion calculation, meal variety, cultural compatibility, and maintaining long-term adherence to prescribed diets. These barriers may be especially pronounced for racial and ethnic minority populations, whose chronic disease outcomes are often worse and whose culturally familiar foods may not be reflected in standard dietary recommendations. This project proposes an AI-driven nutrition agent that continuously adapts to clinical dietary goals, personal preferences, and cultural constraints to generate behaviorally feasible, optimized meal plans.

The proposed system uses an agentic interface to interpret users' health goals, dietary restrictions, taste preferences, cultural food practices, and practical constraints, then translates them into concrete recipes, ingredient portions, and shopping plans. Existing tools such as MacroFactor and Noom primarily emphasize tracking and behavioral coaching, while Medical Nutrition Therapy (MNT) offers expert guidance but is constrained by access, cost, and limited appointment time. Although modern LLMs can provide general nutrition advice on their own, they still struggle with quantitative nutritional reasoning. Rather than relying on generic food recommendations, the proposed system uses food and nutrient databases, recipe knowledge, and constrained optimization to generate meals that satisfy dynamic nutritional requirements while remaining culturally familiar and enjoyable. The system aims to reduce the nutrition-literacy and numeracy burden associated with chronic disease management by removing the need for users to manually calculate nutrients, calories, and ingredient substitutions.

By combining an AI-driven nutrition agent with structured nutrition databases and culturally informed recipe modeling, this project aims to make therapeutic nutrition more practical, personalized, and sustainable. Potential outputs include personalized meal plans, adaptive recipe modifications, and dynamically generated shopping lists aligned with users' evolving health conditions, preferences, and cultural eating patterns. By reducing the cognitive burden of planning meals under complex dietary constraints, the system seeks to improve long-term dietary adherence and help translate clinical nutrition guidance into realistic daily eating habits.

Links:

1. Relevant papers, market research
 - [Interventions to enhance adherence to dietary advice for preventing and managing chronic diseases in adults](#)
 - [Adherence in Chronic Disease](#)
 - [Nutrition literacy predicts adherence to healthy/unhealthy diet patterns in adults with a nutrition-related chronic condition](#)
 - [Addressing Health Literacy and Numeracy to Improve Diabetes Education and Care](#)
 - [Diet Adherence among Adults with Type 2 Diabetes Mellitus: A Concept Analysis](#)

- [Exploring the barriers and facilitators of dietary self-care for type 2 diabetes: a qualitative study in Ghana](#)
- [Attitudes and barriers to dietary advice aimed at reducing risk of type 2 diabetes in first-degree relatives of patients with type 2 diabetes](#)
- [Dietary adherence among persons with type 2 diabetes: A concurrent mixed methods study](#)
- [Cultural competence training of dietitians: development and preliminary evaluation](#)
- [Exploring the barriers of adherence to dietary recommendations among patients with type 2 diabetes: A qualitative study in Iran](#)
- [Is ChatGPT an Effective Tool for Providing Dietary Advice?](#)
- 2. Datasets (publicly available datasets)
 - Recipes:
 - [Open Recipes Project](#)
 - [RecipeNLG](#)
 - Nutrition:
 - [USDA FoodData Central](#)
 - Food Density:
 - [FAO/INFOODS UN Food Density Dataset](#)
 - [USDA Food Density Dataset](#)
- 3. Possible data science techniques and why?
 - Convex Optimization
 - Nutrition databases contain structured numeric nutrient information for foods (protein, fiber, sodium, Vitamin C, etc.).
 - Ingredient portion selection under nutrient targets and dietary constraints can be formulated as a convex optimization problem.
 - Supports macronutrient balancing, sodium/calorie limits, ingredient-ratio constraints, and dietary restriction feasibility.
 - Constraint-based recommender systems
 - Allows recommendations informed by medical constraints, cultural preferences, ingredient exclusions, and behavioral feasibility constraints
 - Relevant Literature
 - [Recommender Systems Handbook](#)
 - [Recommender Systems Textbook](#)
 - [Constraint-based recommender systems: technologies and research issues](#)
 - Agentic/RAG Systems
 - Use LLMs as an interface layer to interpret user intent, identify dietary constraints, and understand cultural preferences
 - Agent tools will be a combination of ingredient ratio models, nutrition database lookups, semantic searches of recipe/ingredient text data, and more.

Link to Business model canvas [optional, mandatory for product oriented projects]: [Business Model Canvas GDrive PDF Link](#)